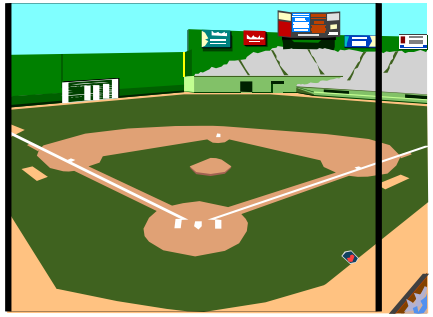
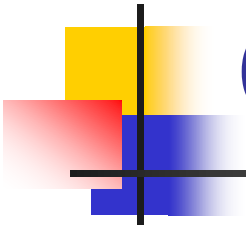




PROJECT MANAGEMENT





Outline

- What is a “project”?
- Project Management
- Objectives and tradeoffs
- Planning and Control in Projects
- Scheduling Methods
- Constant-Time Networks
- PERT and CPM Methods
- Use of Project Concepts

What is a "Project"?

- A project is a series of related jobs directed towards some major output and requiring a significant period of time to perform.
- Distinguishing characteristics of projects
- Examples – New product introduction
  Installation of Equipment

What is “project management”?

- Project management can be defined as

  planning, directing, and controlling resources (i.e., people, equipment, material) to meet the technical, cost and time constraints of the project.

Examples

Building construction	Designing an advertising campaign
Research and development	Startup and shutdown of a plant
Computer systems design	Manufacturing of aircraft, ships or large machines
Fund raising	Planning a military invasion

Objectives and tradeoffs

- Three objectives:-

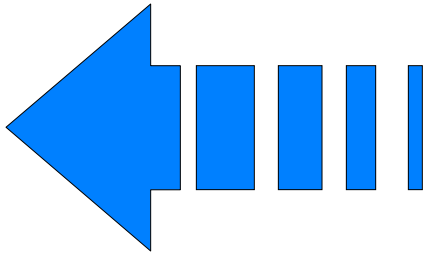
Costs- Sum of direct and allocated costs

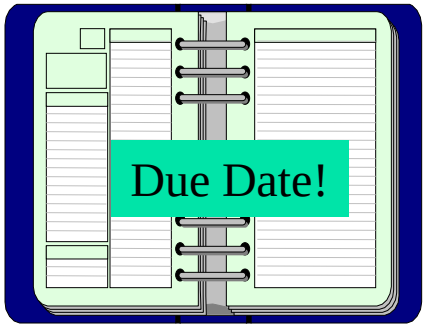
Schedule – Project milestones and completion dates are set at the outset. Meet them or trade-off between costs

Performance – the performance characteristics of the product or service. Trade-off may also be required here.

Objectives and Tradeoffs







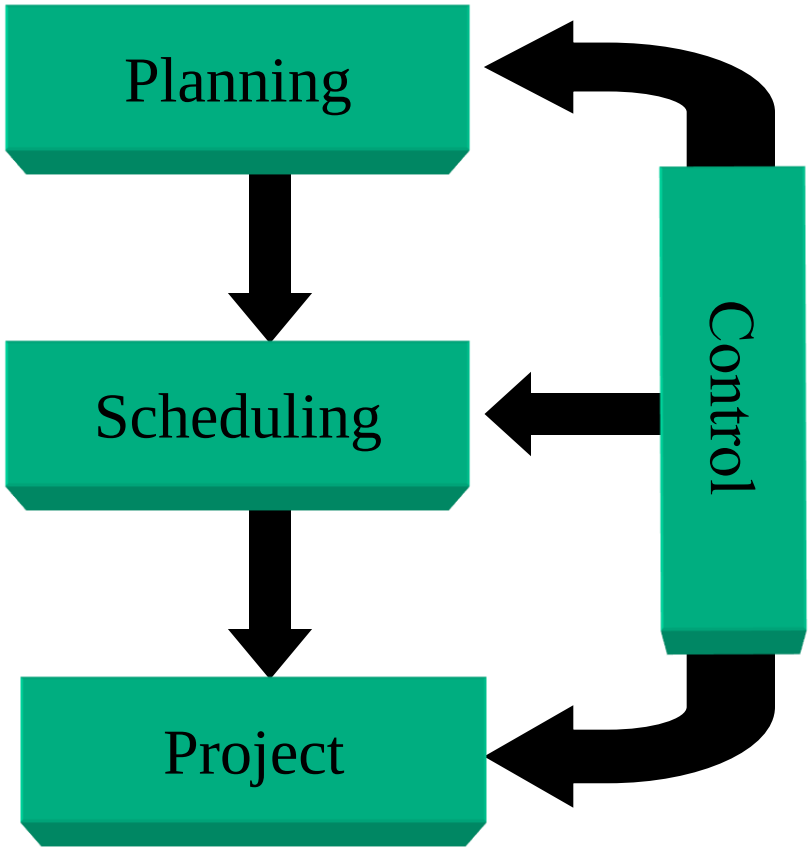
Stay within the budget

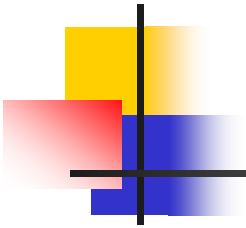


Meet the deadline

Meet the specifications

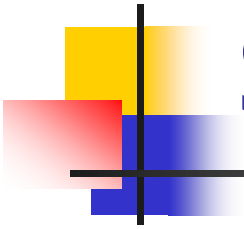
Project Management Activities and Decisions





Planning

- Identify the project customer
- Establish the end product or service
- Set project objectives
- Estimate total resources and time required
- Decide on the form of project organization
- Make key personnel appointments
- Define major tasks required
- Establish a budget



Scheduling

- Develop a detailed work breakdown structure
- Estimate time required for each task
- Sequence the task in the proper order
- Develop a start/stop time for each task
- Develop a detailed budget for each task
- Assign people to tasks

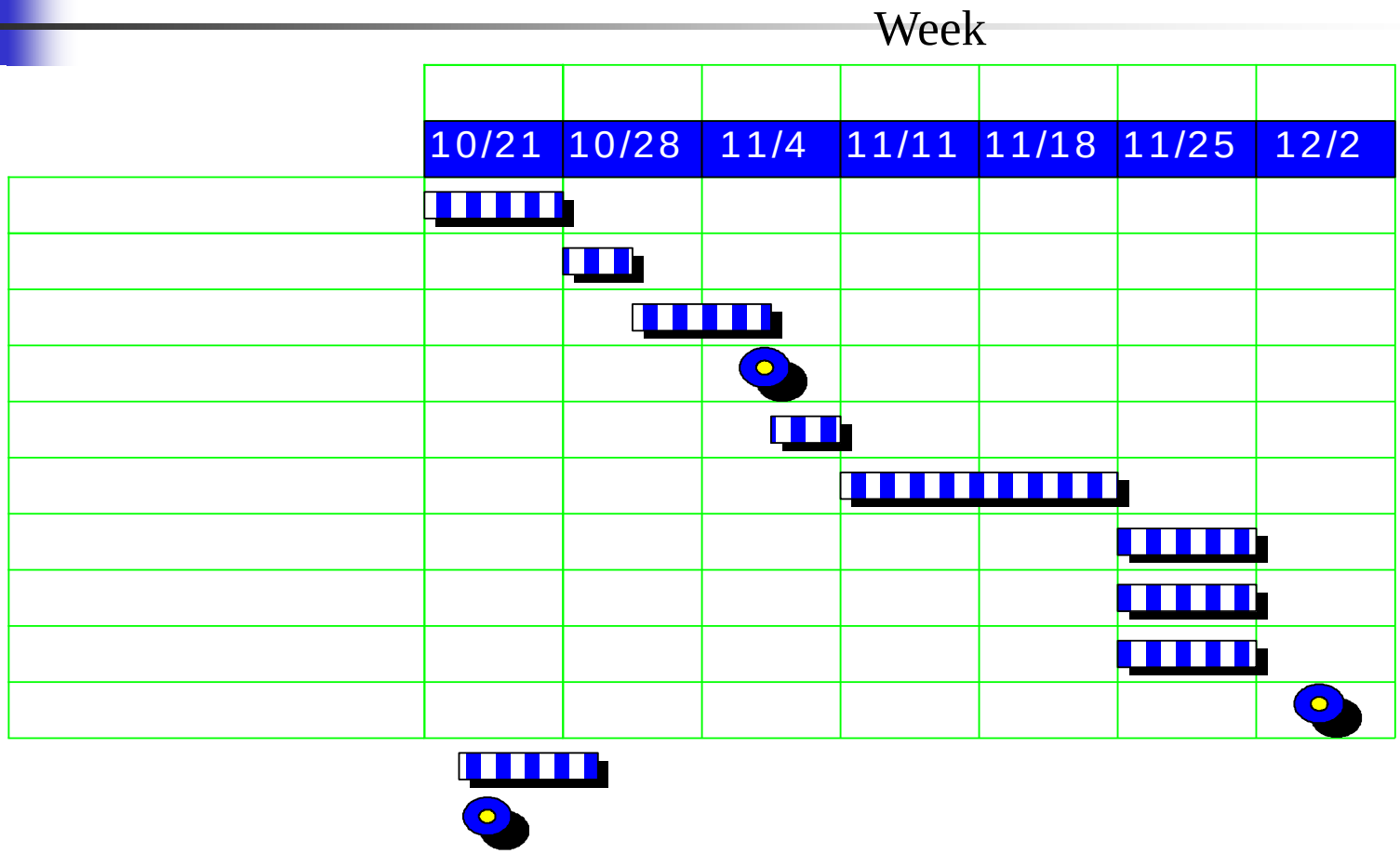
Control

- Monitor actual time, cost, and performance
- Compare planned to actual figures
- Determine whether corrective action is needed
- Evaluate alternative corrective actions
- Take appropriate corrective action

Scheduling Methods

- Two main types of scheduling methods are in use. They are generally classified as
- 1. Gantt charts
- 2. Network Methods
    a. CPM
    b. PERT

Gantt Chart Project Example (Figure 1)



Network Methods-1

- CPM – Critical Path Method
- Developed by E.I. Duponds & Co as a way to schedule the startup and shutdown of major plants.
- Plant activities were repeated often so activity times were fairly well known.
- CPM uses time-cost trade-off.
- Four estimates for each activity are used: normal time, normal cost, crash time and crash cost

Activity	*Description*	*Predecessor*	*Duration in Days*
A	Layout sensing station	None	5
B	Procure rain gauge/wind sensor	None	25
C	Procure telemetry equipment	None	18
D	Construct first 50% of building	A	14
E	Construct remainder of building	D	14
F	Install rain gauge/wind sensor	B, D	4
G	Install telemetry equipment	C, D	5
H	Connect telemetry equipment to rain gauge/wind sensor	F, G	3
I	Final test	E, H	1

CPM Network Diagram - AOA

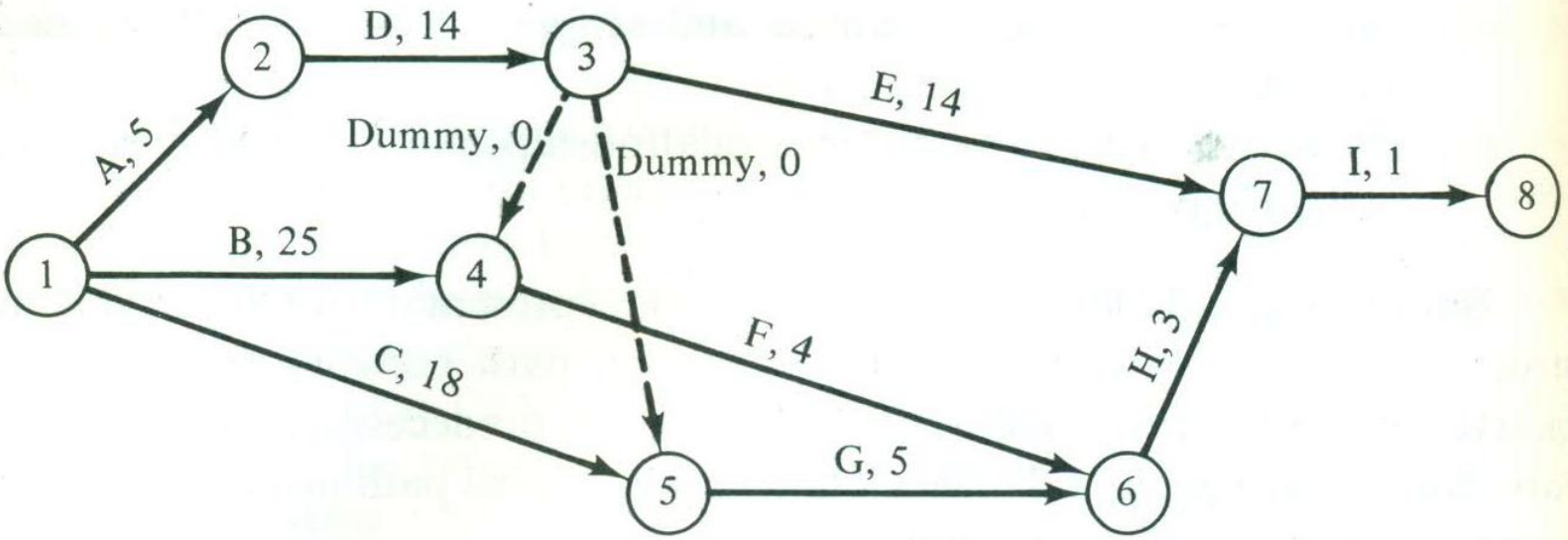
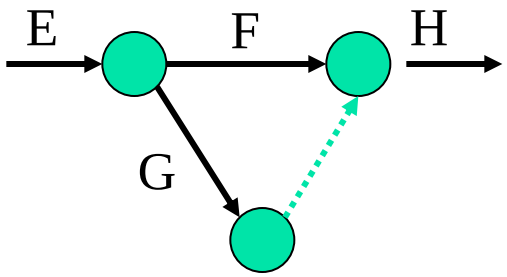


Fig. 17.1. Critical path network of rain gauge/wind sensor installation.

→ An arrow represents one of the *activities* of the project. Its length is of no significance.

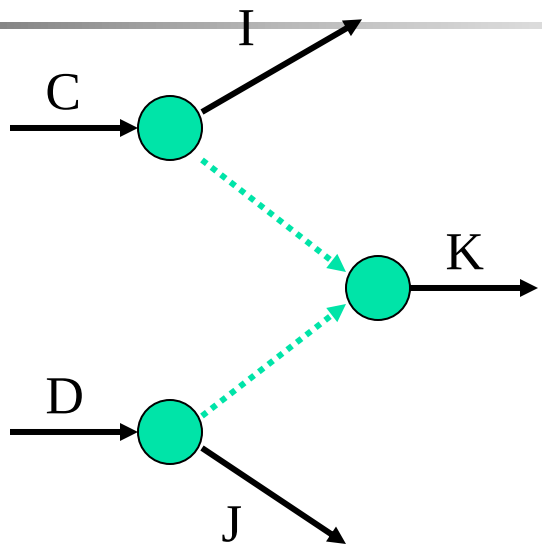
– – → A dashed arrow represents a *dummy activity*. Dummy activities are used to represent precedence relationships. They are not activities in the real sense and have a duration of zero.

Dummy Activity Situations (Figure 4a)



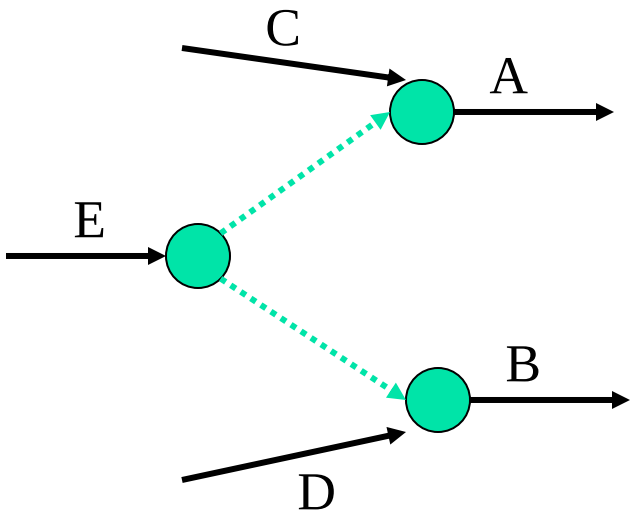
a. Activities F and G have
the same predecessor (E)
and the same successor (H)

Dummy Activity Situations (Figure 4c)



c. Activities C and D have a common successor (K), but also have different successors (I and J)

Dummy Activity Situations (Figure 4b)



b. Activity A and B have a common predecessor (E), but they also have different predecessors (C and D)

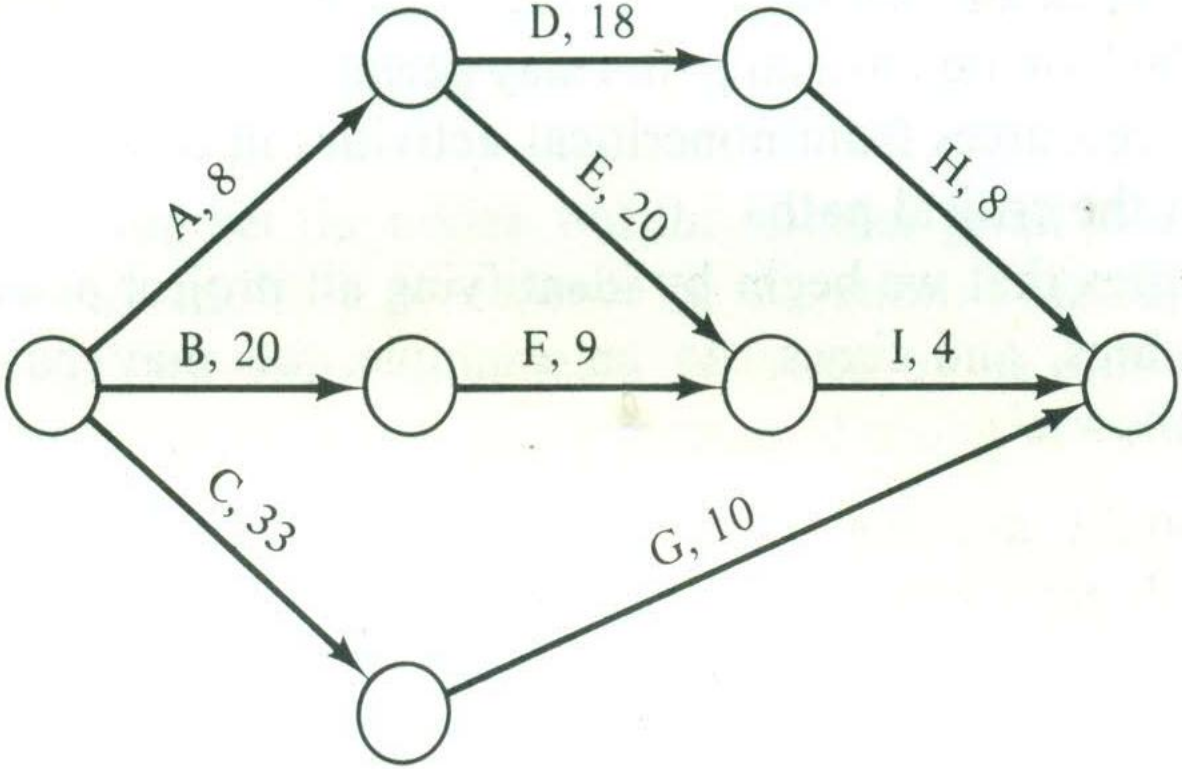
ACTIVITY DURATION

Suppose that we estimate the activity durations for our example as follows:

A:	8 days	F:	9 days
B:	20 days	G:	10 days
C:	33 days	H:	8 days
D:	18 days	I:	4 days
E:	20 days		

NETWORK DIAGRAM WITH ACTIVITY DURATION (AOA)

The CPM network for our project can now be illustrated as in Figure 17.2.



Constant-Time Networks

- Activity times assumed to be constant
- "*Dummy*" activities – Duration time of 0 and are sometimes needed to indicate precedence relationships only in drawing networks
- Notations used in calculating start and finish times:
  - ES(a) = Early Start of activity a
  - EF(a) = Early Finish of activity a
  - LS(a) = Late Start of activity a
  - LF(a) = Late Finish of activity a

Calculating ES, EF, LS, LF, and Completion Time

Forward Pass:

ES (a) = 0 for the starting activity

EF (a) = ES (a) + *t* (a)*

ES (a) = max [EF (all predecessors of a)]

Project completion time = max [EF(all ending activities)]

Backward pass:

LF (a) = min [LS (all successors of a)]

LS (a) = LF - *t*(a)*

** t (a) denotes the duration of activity a*

FORWARD PASS

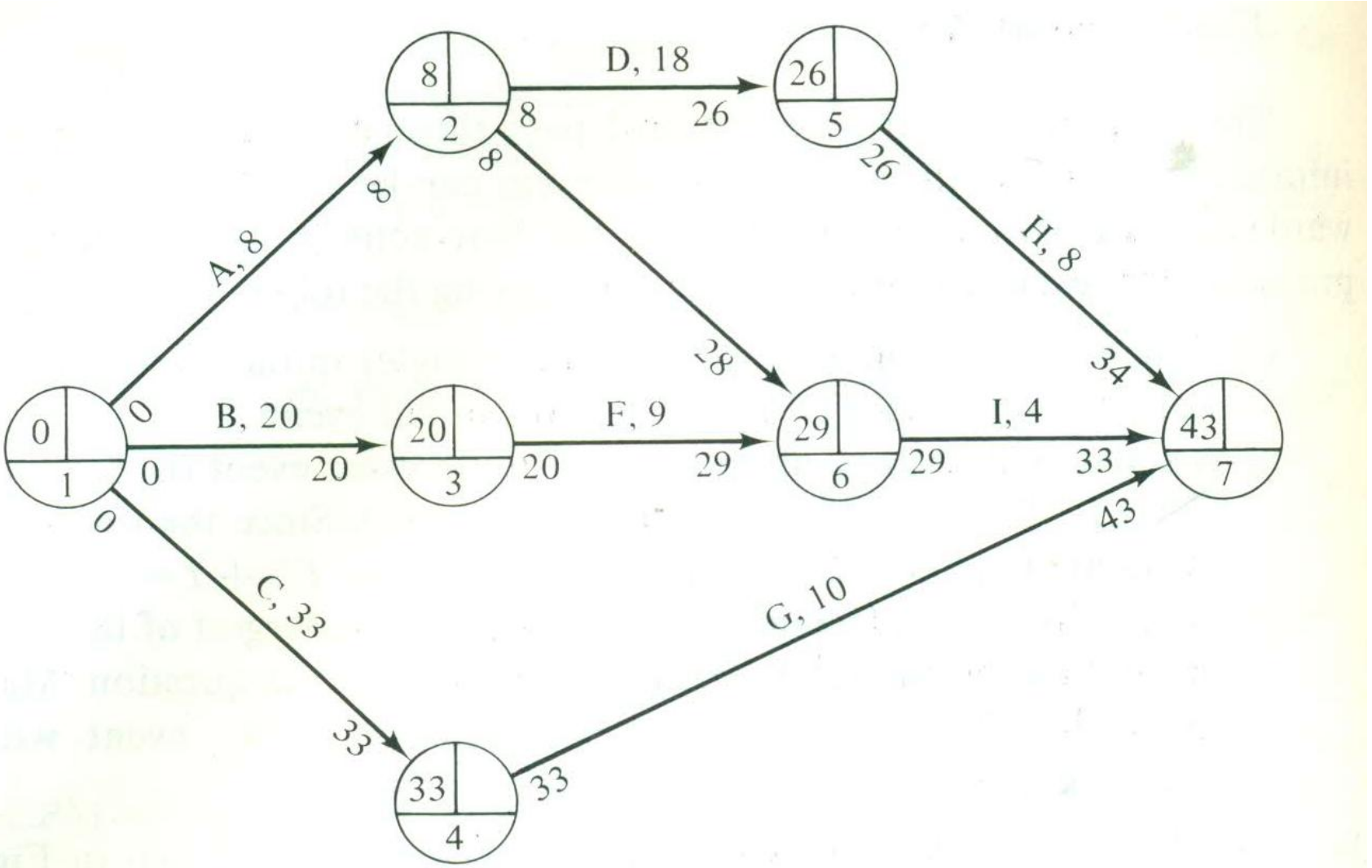


Fig. 17.4. Forward pass calculations.

BACKWARD PASS

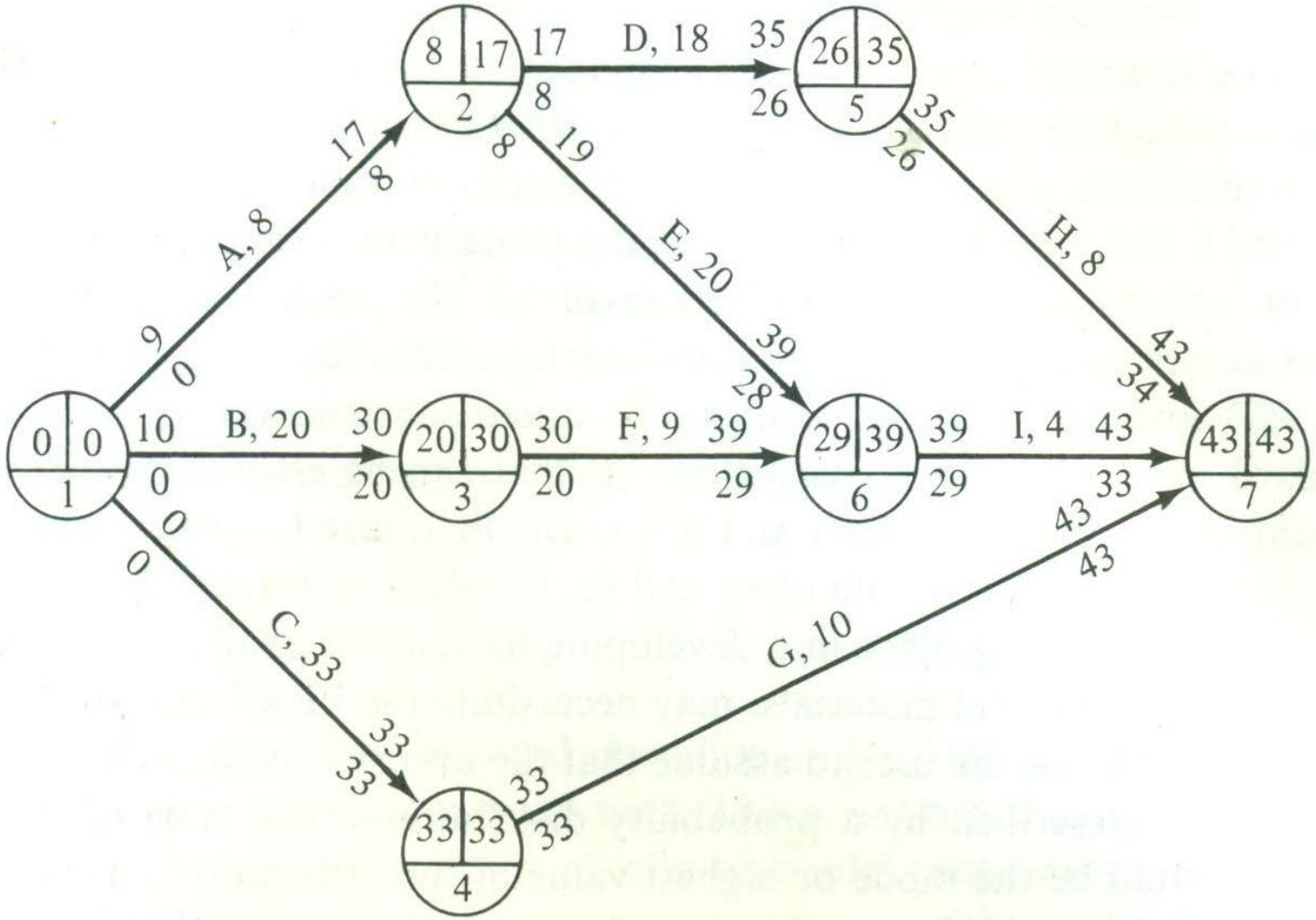


Fig. 17.6. Forward and backward pass calculations.

Forward and Backward Passes

- Information from computations on the networks are used for several managerial purposes.
  1. critical path identification
  2. slack times calculation
  3. Determination of final completion date of project

CRTTICAL PATH

- The critical path is the longest time path through the project network
- It determines project completion
- Activities on the critical path are carefully monitored

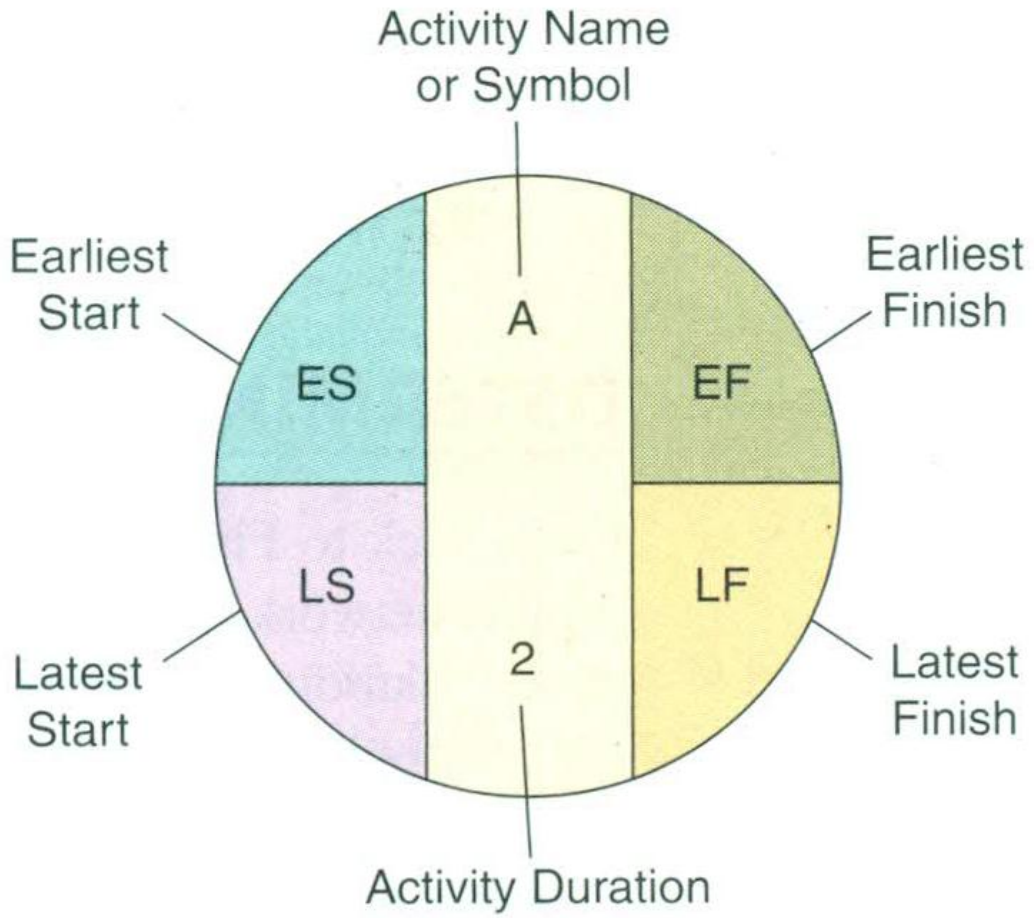
Slack Times

- Total Slack Time = LS (a) - ES (a) = LF (a)- EF (a)
- total slack is the amount of time that an activity duration can be increased without delaying the project completion.
- activities on the critical path will have a slack of zero

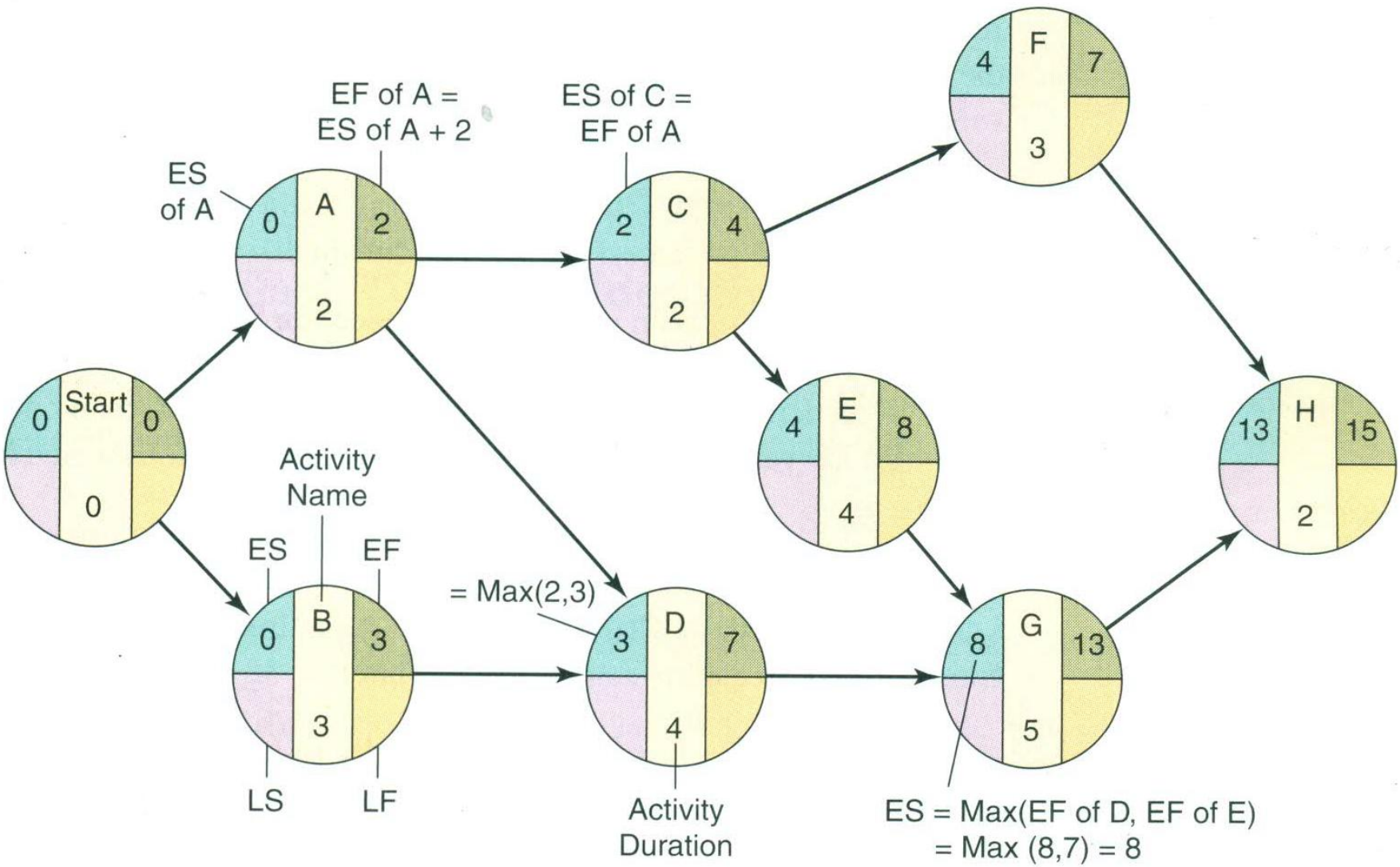
Time Estimates for Milwaukee Paper Manufacturing

ACTIVITY	DESCRIPTION	TIME (WEEKS)
A	Build internal components	2
B	Modify roof and floor	3
C	Construct collection stack	2
D	Pour concrete and install frame	4
E	Build high-temperature burner	4
F	Install pollution control system	3
G	Install air pollution device	5
H	Inspect and test	2
	Total time (weeks)	25

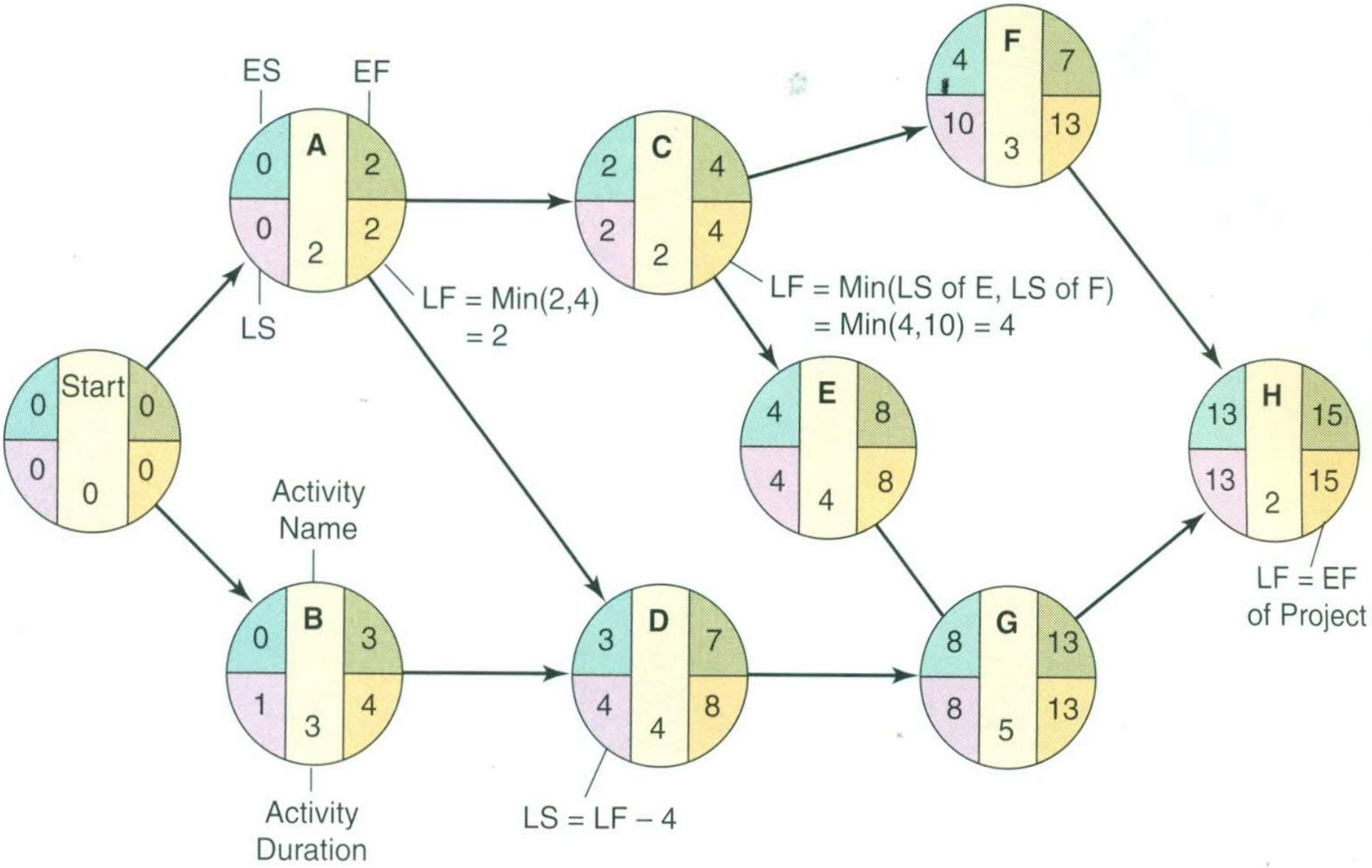
Notation Used in Nodes for Forward and Backward Pass



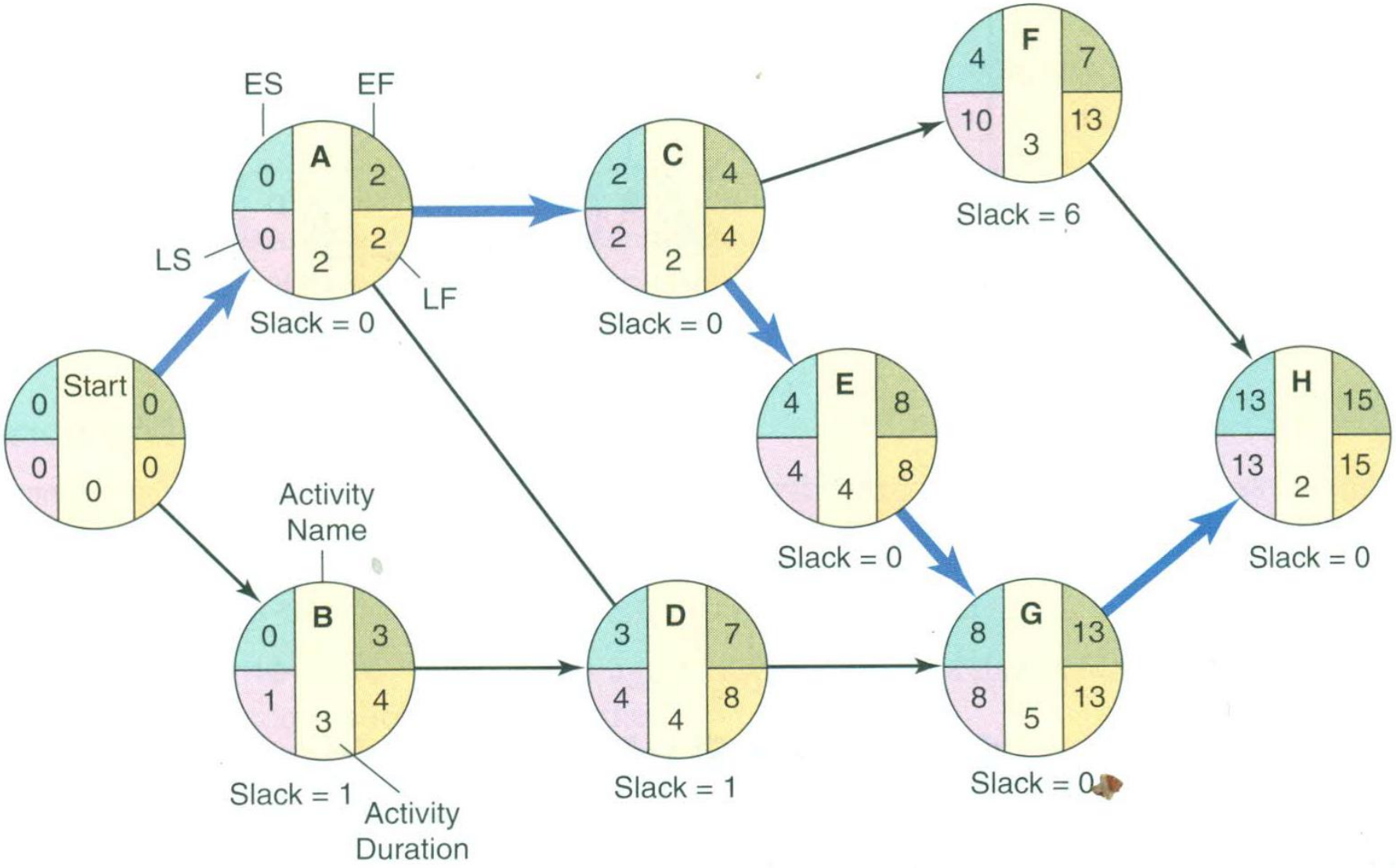
FORWARD PASS



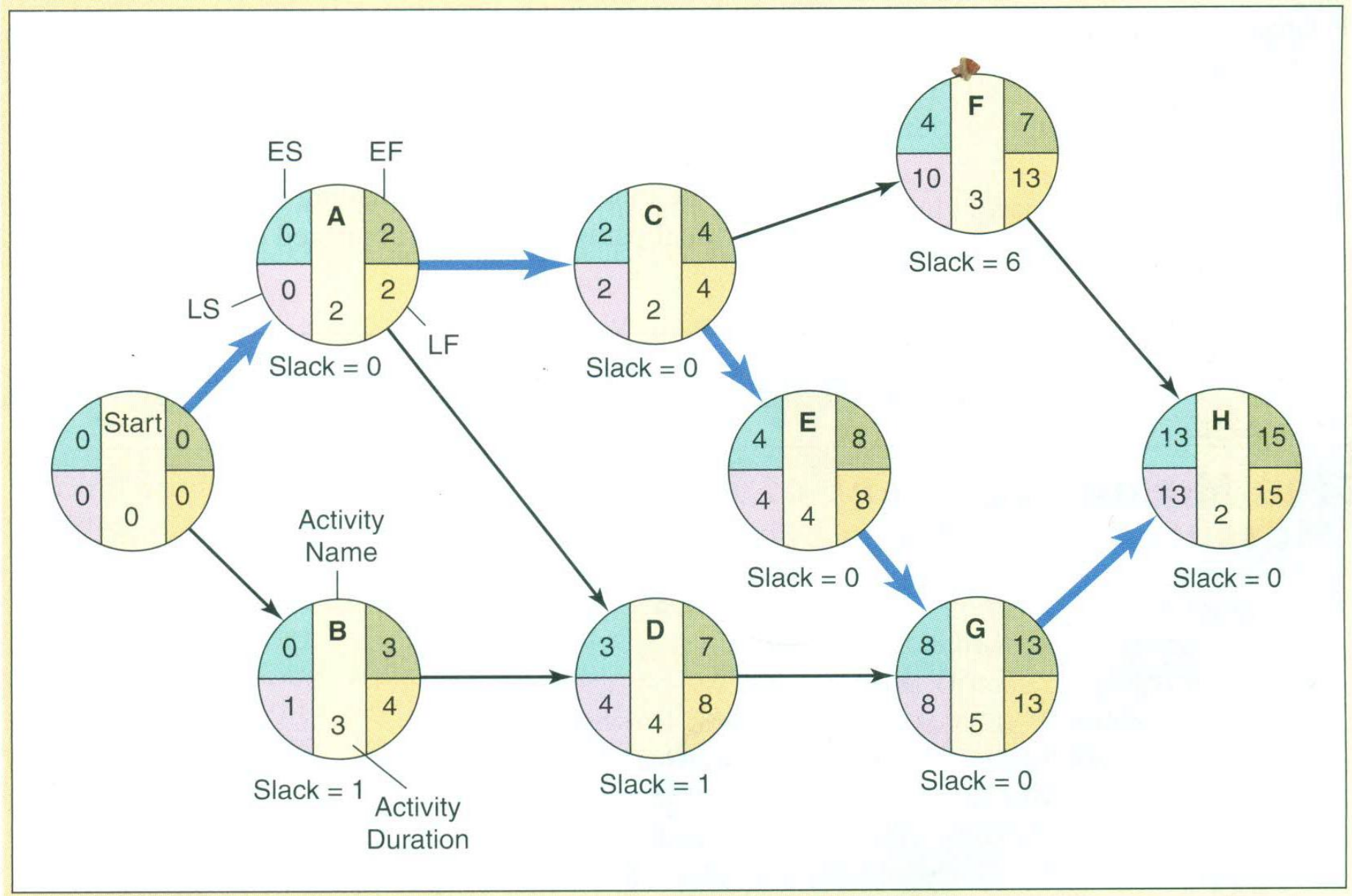
BACKWARD PASS



CRITICAL PATH



CRITICAL PATH and SLACK TIMES



CRASHING

Crashing a project involves four steps, as follows:

Step 1: Compute the crash cost per week (or other time period) for each activity in the network. If crash costs are linear over time, the following formula can be used:

$$\text{Crash cost per period} = \frac{(\text{Crash cost} - \text{Normal cost})}{(\text{Normal time} - \text{Crash time})} \qquad (3\text{-}1$$

Step 2: Using the current activity times, find the critical path(s) in the project network. Identify the critical activities.

Step 3: If there is only one critical path, then select the activity on this critical path that (a) can still b crashed and (b) has the smallest crash cost per period. Crash this activity by one period.

If there is more than one critical path, then select one activity from each critical path such that (a) each selected activity can still be crashed and (b) the total crash cost per period of *all* selected activities is the smallest. Crash each activity by one period. Note that the same activity may be common to more than one critical path.

Step 4: Update all activity times. If the desired due date has been reached, stop. If not, return to Step

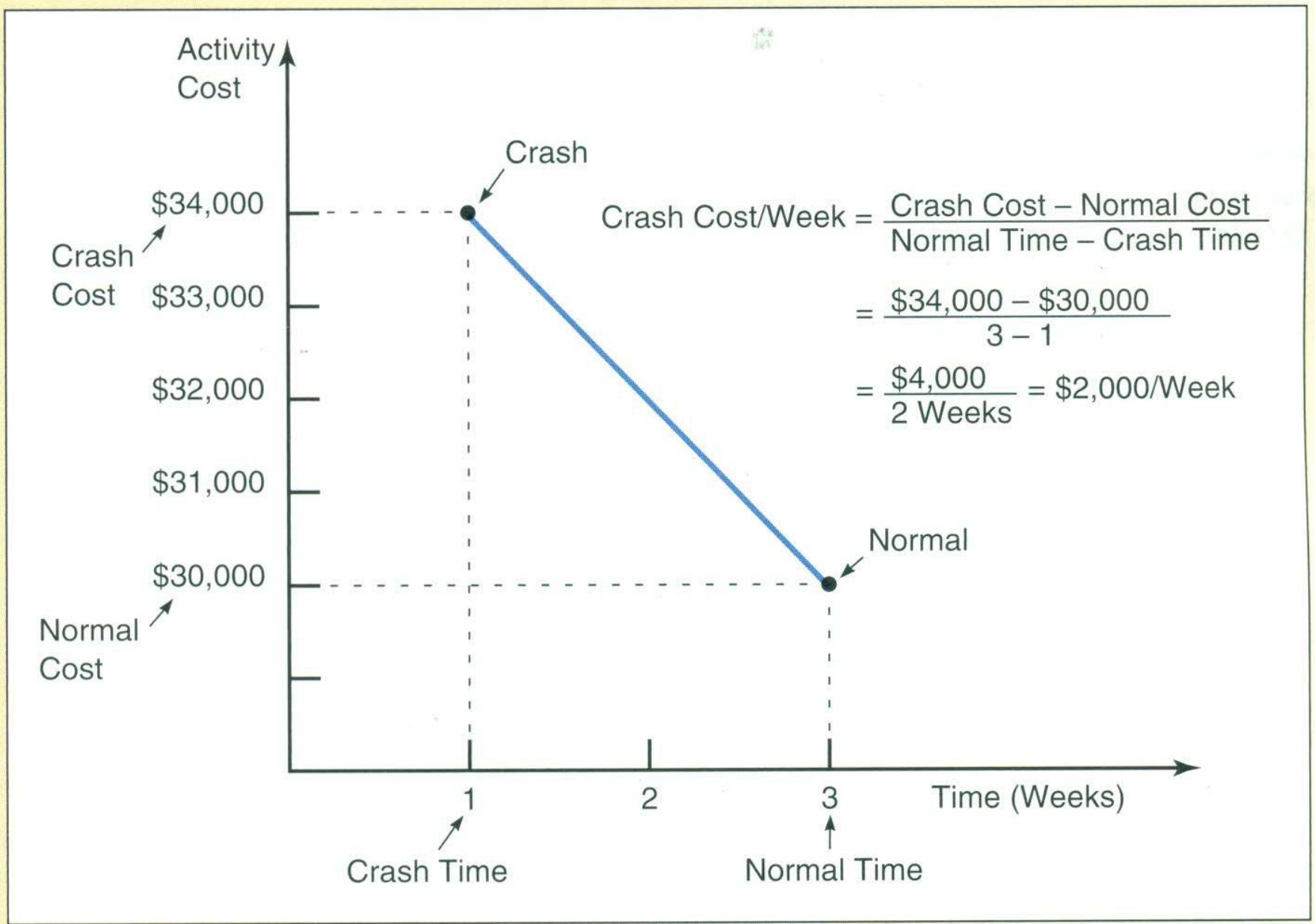
CRASHING

Suppose that Milwaukee Paper Manufacturing has been given only 13 weeks (instead of 16 weeks) to install the new pollution control equipment or face a court-ordered shutdown. As you recall, the length of Joni Steinberg's critical path was 15 weeks. Which activities should Steinberg crash, and by how much, to meet this 13-week due date? Naturally, Steinberg is interested in speeding up the project by 2 weeks, at the least additional cost.

The company's normal and crash times, and normal and crash costs, are shown in Table 3.5. Note, for example, that activity B's normal time is 3 weeks (the estimate used in computing the critical path), and its crash time is 1 week. This means that activity B can be shortened by up to 2 weeks if extra resources are provided. The cost of these additional resources is $4,000 (= difference between the crash cost of $34,000 and the normal cost of $30,000). If we assume that the crashing cost is linear over time (i.e., the cost is the same each week), activity B's crash cost per week is $2,000 (= $4,000/2).

Normal and Crash Data for Milwaukee Paper Manufacturing

	Time (Weeks)		Cost ($)			
Activity	Normal	Crash	Normal	Crash	Crash Cost Per Week ($)	Critical Path?
A	2	1	22,000	22,750	750	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	3	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	84,500	1,500	Yes
H	2	1	16,000	19,000	3,000	Yes


Activity Cost
Crash
$34,000
Crash Cost
$33,000
$32,000
$31,000
$30,000
Normal Cost
Normal
Crash Cost/Week = Crash Cost – Normal Cost / Normal Time – Crash Time
= $34,000 – $30,000 / 3 – 1
= $4,000 / 2 Weeks = $2,000/Week
1
2
3
Time (Weeks)
Crash Time
Normal Time

Network Methods-2

- PERT-Program Evaluation and Review Technique
- First developed in the mid 1950's for the Polaris submarine project. It was used to schedule 3000 subcontractors, suppliers and agencies.
- It is credited with bringing that project ahead of schedule by up to two years.

PERT

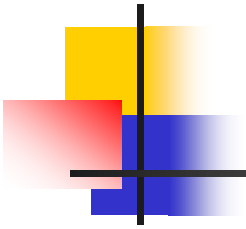
- PERT requires three time estimates for each activity:

  optimistic time estimate

  most likely time estimate

  pessimistic time estimate

  These estimates recognize the uncertainty in activity time for real life situations



PERT

- Activity times are converted to expected time of completion and variance for each activity

Expected Time Calculation

t_o = optimistic time

t_m = most likely time

t_p = pessimistic time

t_e = expected (average) time

The first three times can be combined in a weighted average to determine an expected time. This weighted average is

$$t_e = \frac{t_o + 4t_m + t_p}{6} \qquad (17.1)$$

EXAMPLE OF CALCULATION

Suppose Joni Steinberg and the project management team at Milwaukee Paper developed the following time estimates for Activity F (Installing the Pollution Control System):

$$a = 1 \text{ week}, \; m = 2 \text{ weeks}, \; b = 9 \text{ weeks}$$

a. Find the expected time and variance for Activity F.

b. Then compute the expected time and variance for all of the other activities in the pollution control project. Use the time estimates in Table 3.4.

SOLUTION

a. The expected time for Activity F is

$$t = \frac{a + 4m + b}{6} = \frac{1 + 4(2) + 9}{6} = \frac{18}{6} = 3 \text{ weeks}$$

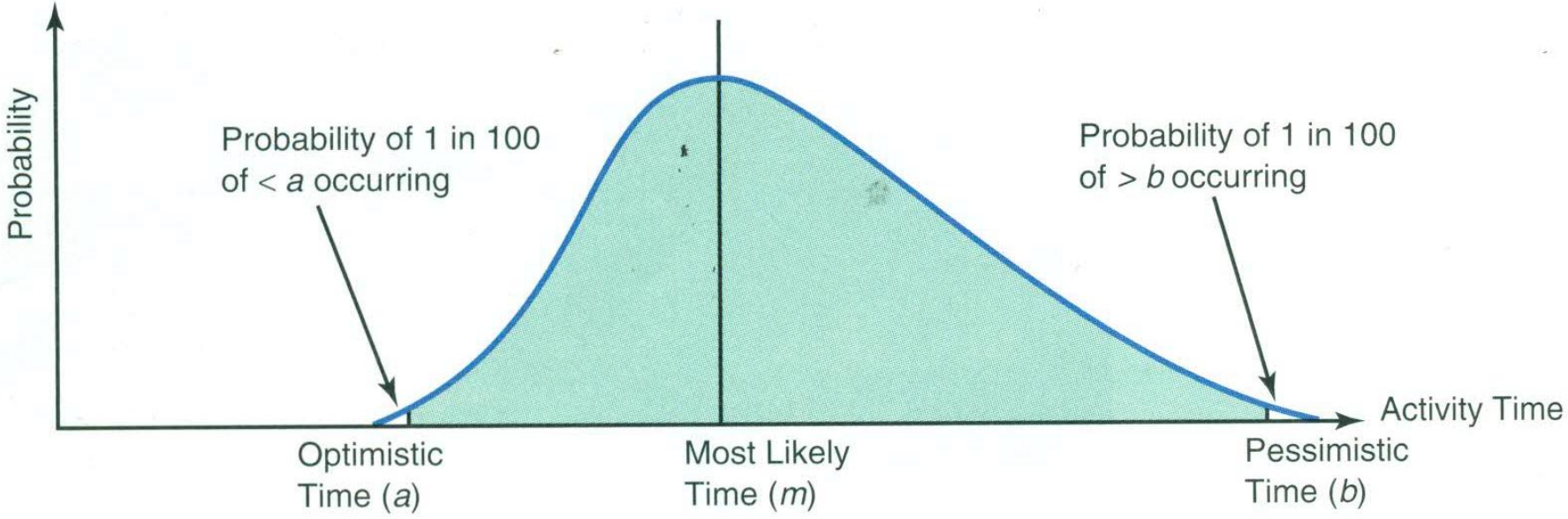
The variance for Activity F is

$$\text{Variance} = \left[\frac{(b - a)}{6}\right]^2 = \left[\frac{(9 - 1)}{6}\right]^2 = \left(\frac{8}{6}\right)^2 = \frac{64}{36} = 1.78$$

Time Estimates (in weeks) for Milwaukee Paper's Project

Activity	Optimistic a	Most Likely m	Pessimistic b	Expected Time $t = (a + 4m + b)/6$	Variance $[(b - a)/6]^2$
A	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$
B	2	3	4	3	$[(4 - 2)/6]^2 = 4/36 = .11$
C	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$
D	2	4	6	4	$[(6 - 2)/6]^2 = 16/36 = .44$
E	1	4	7	4	$[(7 - 1)/6]^2 = 36/36 = 1.00$
F	1	2	9	3	$[(9 - 1)/6]^2 = 64/36 = 1.78$
G	3	4	11	5	$[(11 - 3)/6]^2 = 64/36 = 1.78$
H	1	2	3	2	$[(3 - 1)/6]^2 = 4/36 = .11$

Beta Probability Distribution with Three Time Estimates



To compute the *dispersion* or *variance of activity completion time*, we use the formula:[3]

$$\text{Variance} = [(b - a)/6]^2 \qquad (3\text{-}7$$

Probability of Project Completion

From Example 8 (see Table 3.4), we know that the variance of activity A is 0.11, variance of activity C is 0.11, variance of activity E is 1.00, variance of activity G is 1.78, and variance of activity H is 0.11.

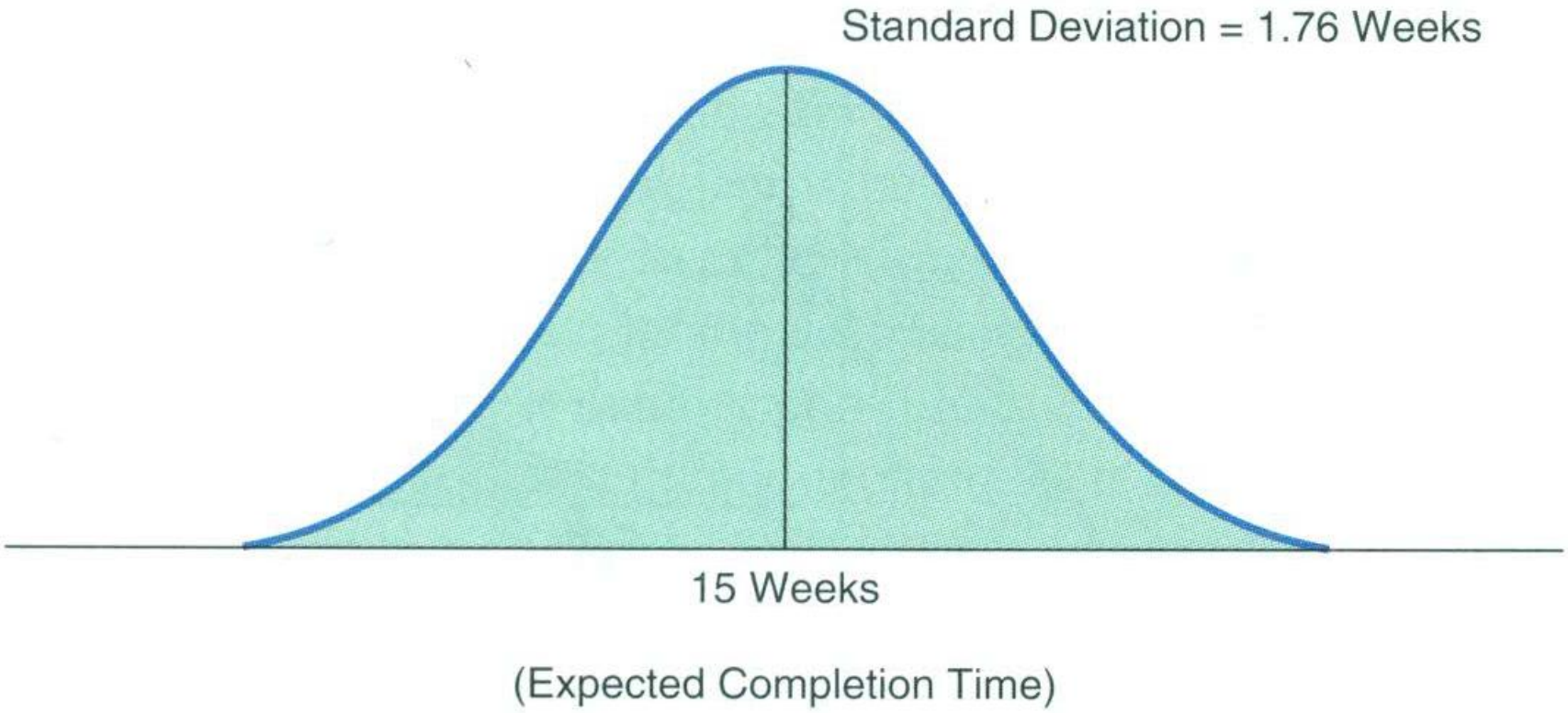
Compute the total project variance and project standard deviation.

$$\text{Project variance } (\sigma_p^2) = 0.11 + 0.11 + 1.00 + 1.78 + 0.11 = 3.11$$

which implies

$$\text{Project standard deviation } (\sigma_p) = \sqrt{\text{Project variance}} = \sqrt{3.11} = 1.76 \text{ weeks}$$

Probability Distribution for Project Completion Times at Milwaukee Paper



Probability of completing a project on-time

Joni Steinberg would like to find the probability that her project will be finished on or before the 16-week deadline.

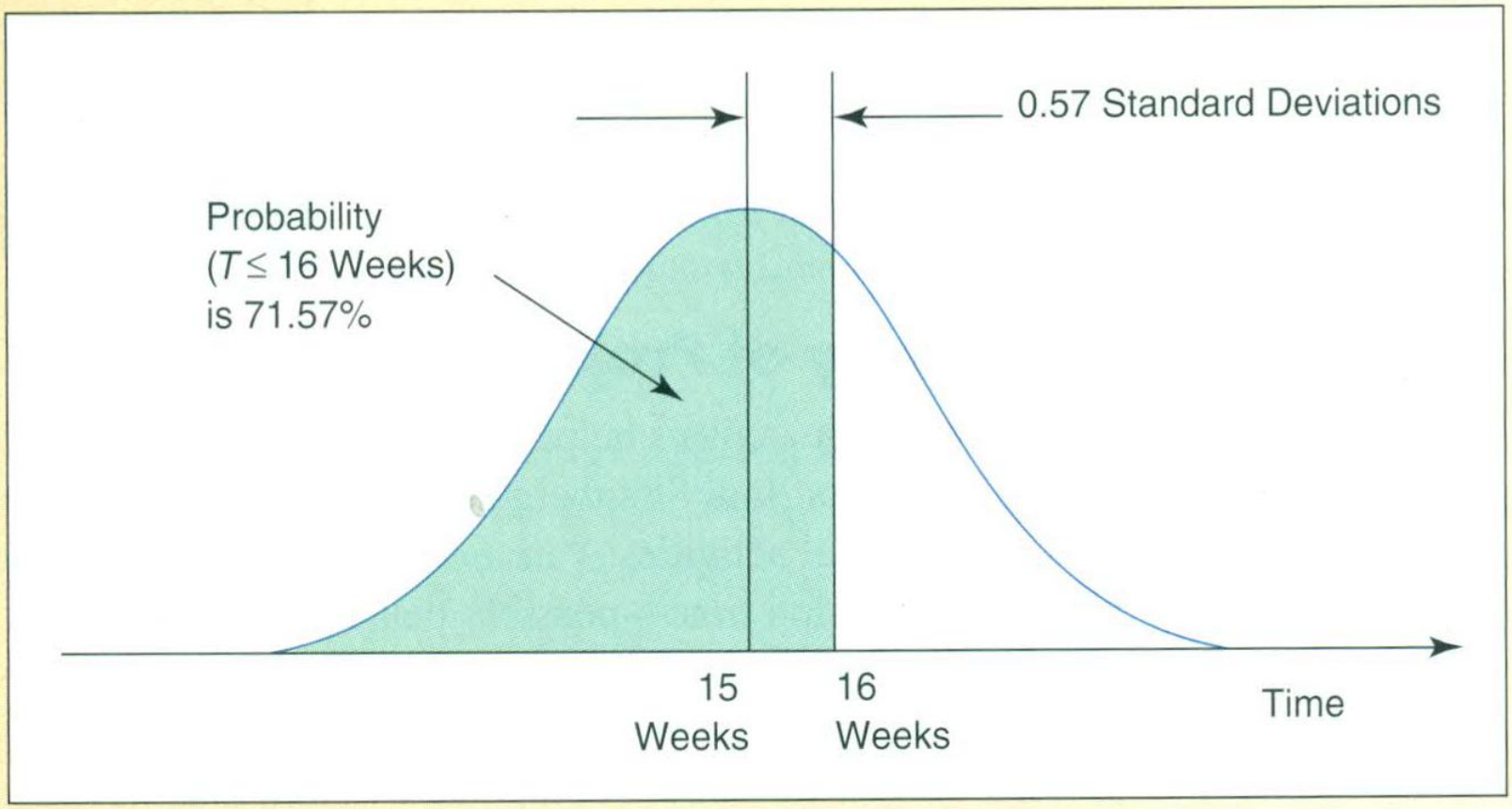
To do so, she needs to determine the appropriate area under the normal curve. The standard normal equation can be applied as follows:

$$Z = (\text{due date} - \text{expected date of completion})/\sigma_p \qquad (3\text{-}9)$$

$$= (16 \text{ weeks} - 15 \text{ weeks})/1.76 \text{ weeks} = 0.57$$

where Z is the number of standard deviations the due date or target date lies from the mean or expected date.

Probability that Milwaukee Paper will Meet the 16-Week Deadline



A. Areas under the normal curve, 0 to z, page 879
B. Areas under the standardized normal curve
1. From $-\infty$ to $-z$, page 880
2. From $-\infty$ to $+z$, page 881
C. Cumulative Poisson probabilities, page 882
D. Cumulative binomial probabilities, page 884

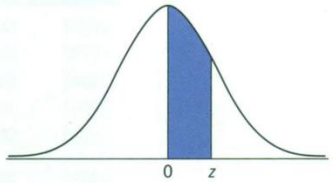
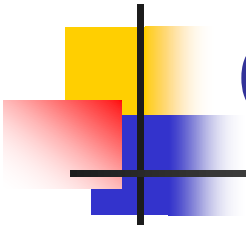


Table A Areas under the normal curve, 0 to z

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	.0000	.0040	.0080	.0120	.0160	.0199	.0239	.0279	.0319	.0359
0.1	.0398	.0438	.0478	.0517	.0557	.0596	.0636	.0675	.0714	.0753
0.2	.0793	.0832	.0871	.0910	.0948	.0987	.1026	.1064	.1103	.1141
0.3	.1179	.1217	.1255	.1293	.1331	.1368	.1406	.1443	.1480	.1517
0.4	.1554	.1591	.1628	.1664	.1700	.1736	.1772	.1808	.1844	.1879
0.5	.1915	.1950	.1985	.2019	.2054	.2088	.2123	.2157	.2190	.2224
0.6	.2257	.2291	.2324	.2357	.2389	.2422	.2454	.2486	.2517	.2549
0.7	.2580	.2611	.2642	.2673	.2703	.2734	.2764	.2794	.2823	.2852
0.8	.2881	.2910	.2939	.2967	.2995	.3023	.3051	.3078	.3106	.3133
0.9	.3159	.3186	.3212	.3238	.3264	.3289	.3315	.3340	.3365	.3389
1.0	.3413	.3438	.3461	.3485	.3508	.3531	.3554	.3577	.3599	.3621
1.1	.3643	.3665	.3686	.3708	.3729	.3749	.3770	.3790	.3810	.3830
1.2	.3849	.3869	.3888	.3907	.3925	.3944	.3962	.3980	.3997	.4015
1.3	.4032	.4049	.4066	.4082	.4099	.4115	.4131	.4147	.4162	.4177
1.4	.4192	.4207	.4222	.4236	.4251	.4265	.4279	.4292	.4306	.4319
1.5	.4332	.4345	.4357	.4370	.4382	.4394	.4406	.4418	.4429	.4441
1.6	.4452	.4463	.4474	.4484	.4495	.4505	.4515	.4525	.4535	.4545
1.7	.4554	.4564	.4573	.4582	.4591	.4599	.4608	.4616	.4625	.4633
1.8	.4641	.4649	.4656	.4664	.4671	.4678	.4686	.4693	.4699	.4706
1.9	.4713	.4719	.4726	.4732	.4738	.4744	.4750	.4756	.4761	.4767
2.0	.4772	.4778	.4783	.4788	.4793	.4798	.4803	.4808	.4812	.4817
2.1	.4821	.4826	.4830	.4834	.4838	.4842	.4846	.4850	.4854	.4857
2.2	.4861	.4864	.4868	.4871	.4875	.4878	.4881	.4884	.4887	.4890
2.3	.4893	.4896	.4898	.4901	.4904	.4906	.4909	.4911	.4913	.4916
2.4	.4918	.4920	.4922	.4925	.4927	.4929	.4931	.4932	.4934	.4936
2.5	.4938	.4940	.4941	.4943	.4945	.4946	.4948	.4949	.4951	.4952
2.6	.4953	.4955	.4956	.4957	.4959	.4960	.4961	.4962	.4963	.4964
2.7	.4965	.4966	.4967	.4968	.4969	.4970	.4971	.4972	.4973	.4974
2.8	.4974	.4975	.4976	.4977	.4977	.4978	.4979	.4979	.4980	.4981
2.9	.4981	.4982	.4982	.4983	.4984	.4984	.4985	.4985	.4986	.4986
3.0	.4987	.4987	.4987	.4988	.4988	.4989	.4989	.4989	.4990	.4990

Use of Project Management Concepts



- Scheduling is only part of a *complete* approach to project management
- Tradeoffs between sophistication and cost of methods
- Choice between PERT and CPM
- Project management software

Project management software

- Microsoft Project
- Primavera Systems Inc.
- Sure Trak 2.0

- Scitor Corp’s Project Scheduler 7
- Kidasa Software, Inc.
- Milestones e.t.c.